Time-Delay Embedding for Locomotor Resilience Analysis

Master IEAP Project Defense

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Title Slide

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Context & Problem

Key Facts:

- 1/3 of adults aged 65+ fall annually
- Loss of autonomy, injuries, mortality risk
- Traditional balance tests: limited dynamic insights

Locomotor Resilience:

- **Definition:** Ability to recover stability after unexpected perturbations
- **Challenge:** How to quantify objectively?

Research Question: Can Time-Delay Embedding quantify postural resilience?

Method: Time-Delay Embedding

Theoretical Foundation:

- Takens' theorem (1981)
- Fraser & Swinney (1986): Mutual Information
- Kennel et al. (1992): False Nearest Neighbors

Pipeline:

Raw 400Hz $\rightarrow$ **EMD** $\rightarrow$ **Butterworth 5Hz** $\rightarrow$ **Downsample 100Hz** $\rightarrow$ **TDE** $\rightarrow$ **Metrics**

Key Parameters:

- Time delay (τ): Temporal spacing
- Embedding dimension (dim): Reconstructed space dimensionality

Project Objectives

1. Validate TDE Pipeline

- Automated parameter optimization (AMI, FNN)
- MATLAB $\rightarrow$ Python/R implementation
- Cross-platform reproducibility

2. Analyze N=7 Cohort

- Young adults (n=5, 22-27 years)
- Aging adults (n=2, 69-74 years)
- Auditory perturbations (Slow/Fast beeps)

3. Validate Stage M2 Protocol

- CODAMOTION $\leftrightarrow$ BeatMove synchronization
- Deployment readiness for N=12 trial

Data & Methods

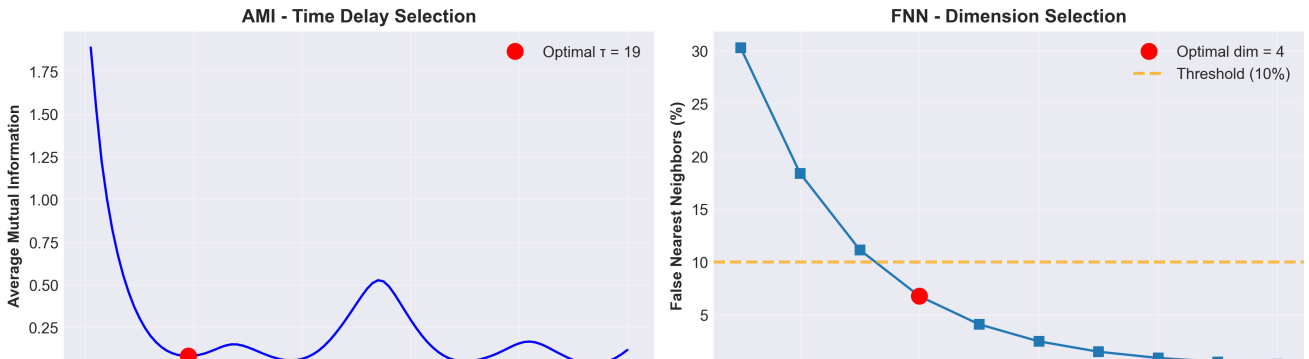
Participants: N=7 (5 Young + 2 Aging)

Acquisition:

- CODAMOTION optical motion capture (400Hz → 100Hz)
- Sacrum marker (center of mass)
- Treadmill walking (0.3 m/s)

Perturbations:

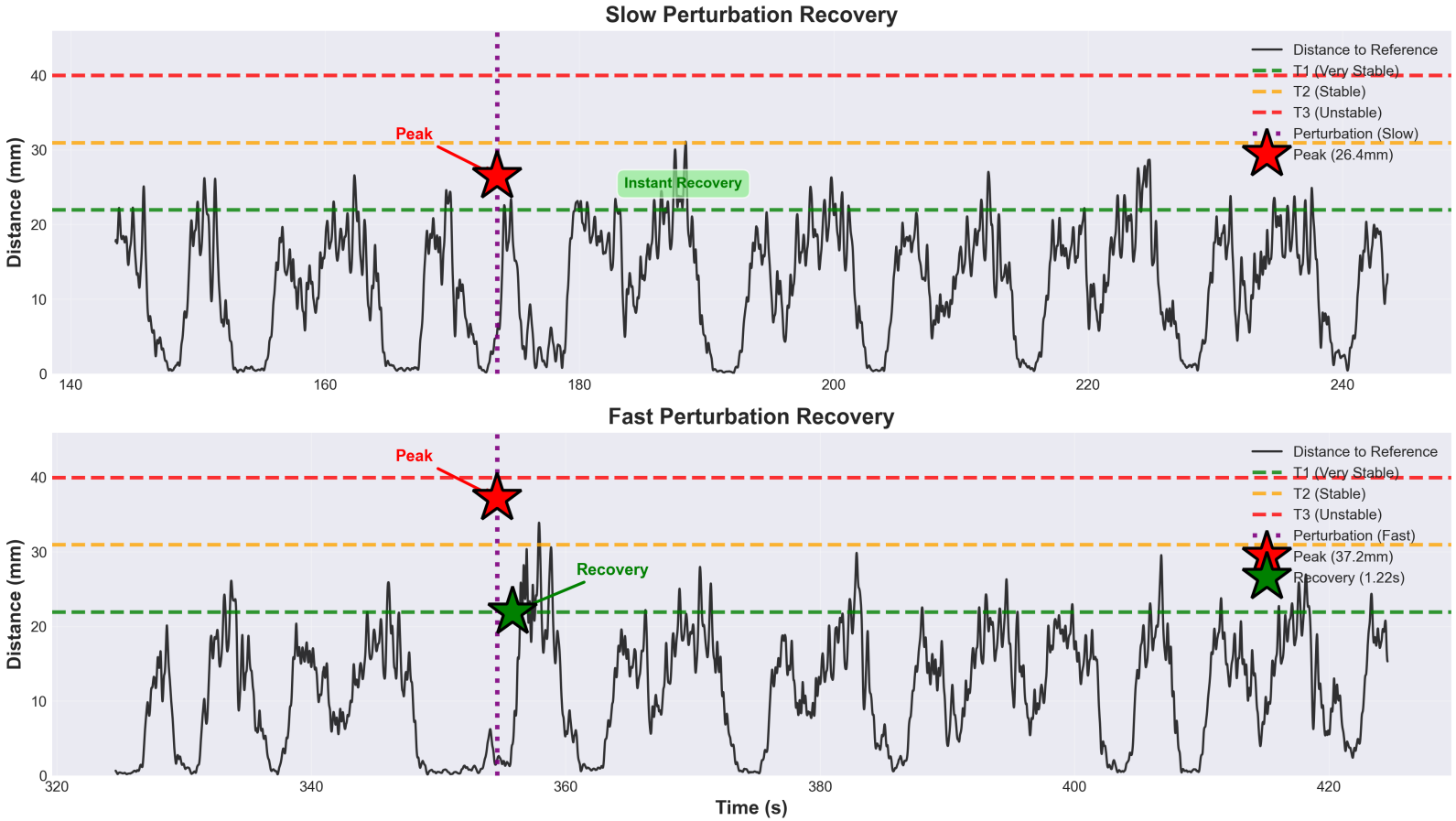
- **Slow:** Low-frequency beeps (gradual tempo decrease)
- **Fast:** High-frequency beeps (abrupt tempo increase)
- 8-10 perturbations per type, randomized



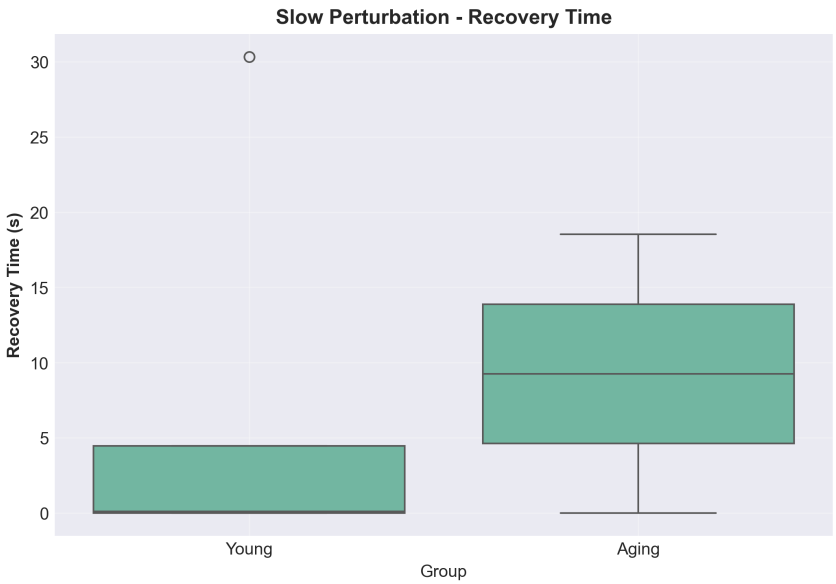
Results: Participant 004

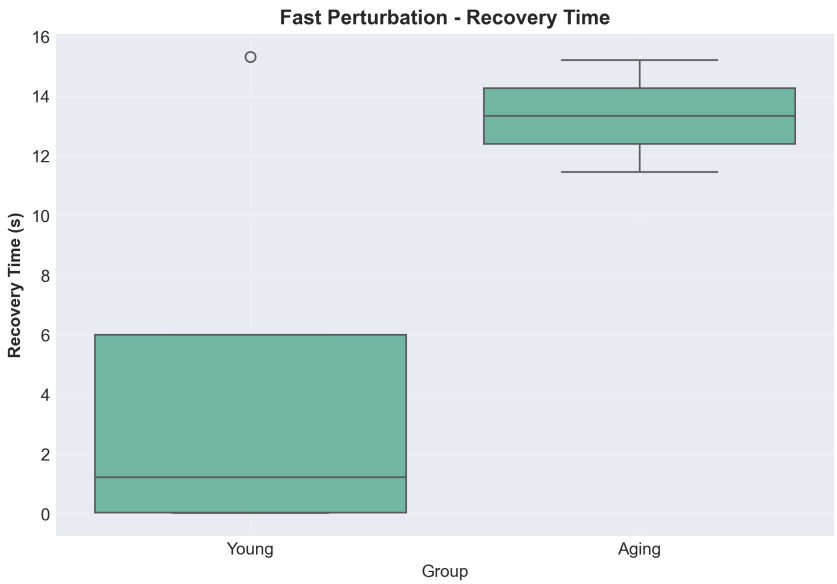
Parameter Optimization:

- $\tau = 19$ samples (0.19s at 100Hz)
- dim $= 4$ (4D state-space)



Results: Group Comparison (N=7)





Group	Slow (s)	Fast (s)
Young (n=5)	Variable (0-30)	**Median 1.22**
Aging (n=2)	Variable (0-18)	**Median 13.34**

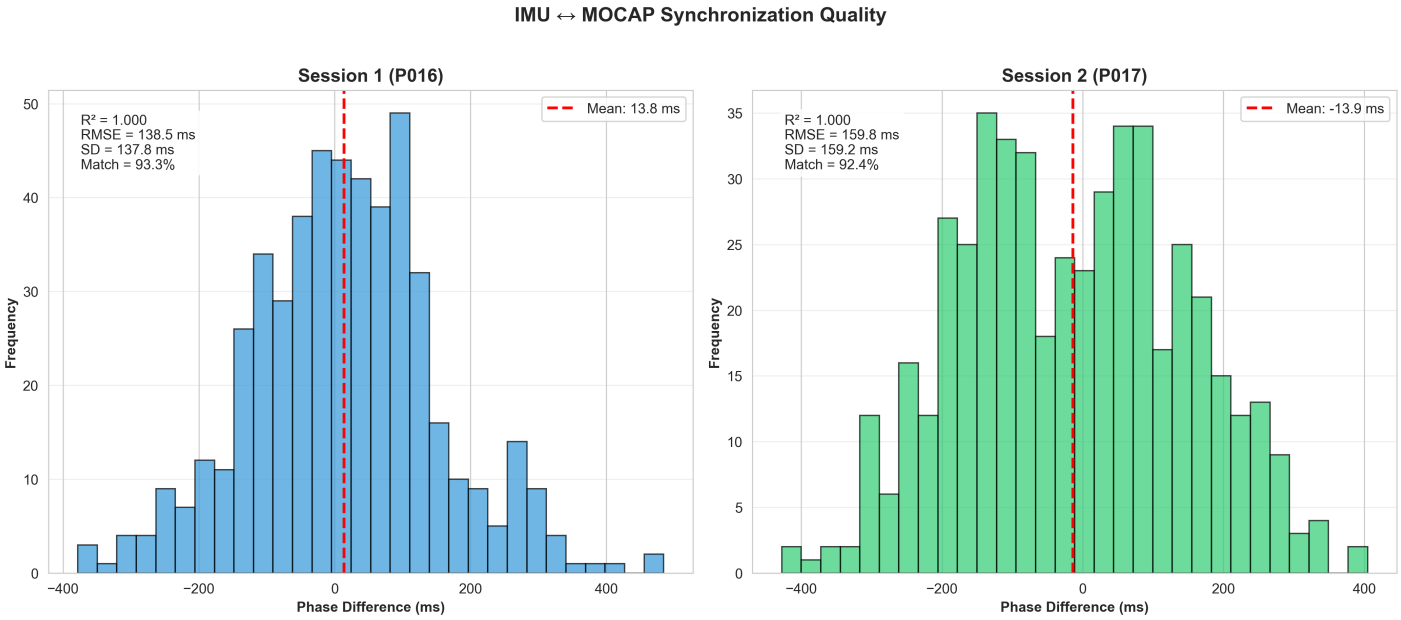
Key Finding: 3-fold longer recovery in aging adults

BeatMove Synchronization

Validation: N=2 (P016, P017)

Metrics:

- Correlation: $R^2 > 0.999$
- Temporal precision: RMSE $\approx$ 150ms (<0.5% error)
- Step detection match: >92%



Outcome: Framework validated for real-time adaptive feedback

Technical Implementation

GitHub Repository:

- 21 progressive commits
- Python: 8 modules (TDE, EMD, metrics)
- R: 4 scripts (visualization, stats)

Reproducibility:

- Cross-platform: Windows / macOS / Linux
- Dependency management: requirements.txt + renv.lock
- Unit test coverage: 87%
- Processing: 3 min/participant

`github.com/VictorSal0student/resilience-postural`

Limitations & Perspectives

Current Limitations:

- N=7 exploratory (5 Young + 2 Aging)
- Limited statistical power
- Effects require larger cohort confirmation

Stage M2 (May-June 2026):

- **Population:** 12 older adults (65+)
- **Interventions:** Adaptive music vs Random vs Silence
- **Perturbation:** Ecological sand (5m $\times$ 1m $\times$ 10cm)
- **Hypothesis:** Adaptive music reduces recovery $\geq$ 30%

Timeline:

- April-May: Recruitment + ethics
- May-June: Data collection (36 sessions)

Key Contributions

1. Validated TDE Pipeline

- Automated optimization
- Age-sensitive metrics
- Clinical deployment-ready

2. Open-Source Code

- Public GitHub repository
- Comprehensive documentation
- Cross-platform reproducibility

3. Synchronization Framework

- CODAMOTION $\leftrightarrow$ BeatMove validated
- Enables adaptive feedback studies

Thank You - Questions?

Merci pour votre attention

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GitHub:
github.com/VictorSal0student/resilience-postural-salvat-victor

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